



Dual VCO module

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Introduction

In this project a voltage-controlled oscillator is designed and build to work with other modules in a modular synthesizer. A voltage-controlled oscillator, or VCO, converts a DC input voltage into an AC output voltage. VCO's are considered to be the heart of any synthesizer, as it mostly makes up the primary signal that can be modulated and outputted as sound.

Objectives

The main objective of this project is to design a dual VCO-module for a modular synthesizer. With the duality being able to output two independent signals.

Requirements

In consultation with the stakeholders, a few requirements have been set up:

Functional requirements:

- The ability to output two independent signals.
- Being able to output three different waveforms.
- The input and output voltages must correspond to the 1V / octave standard.

Non-functional requirements:

- The internal electronics must be easily presentable.
- the linearity of the signal must be adjustable.

Constraints

- The budget of 50 euros.
- The supply voltage of +/- 12 volts.
- The final size of the module.

Research

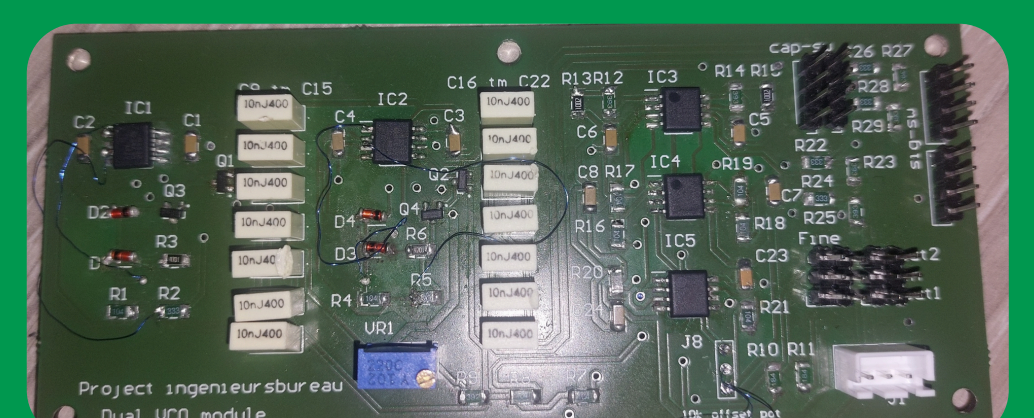
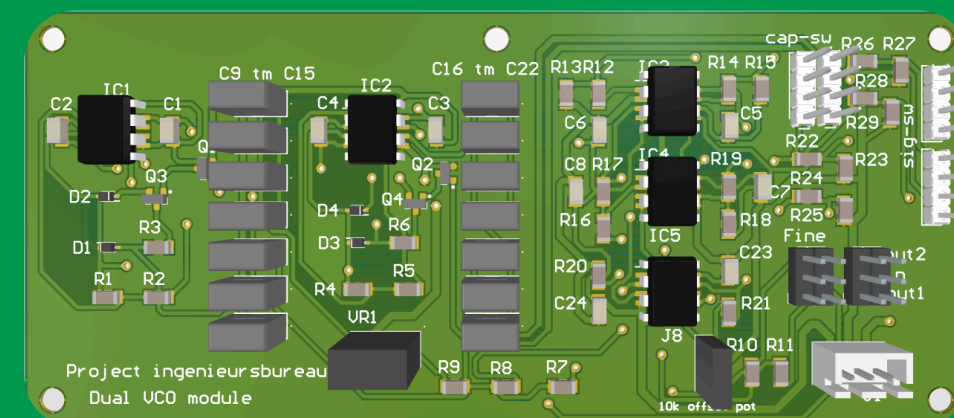
how do you create and design a voltage controlled oscillator used in synthesizers?

Method

To make a VCO module simple it can be split into three different parts. First a simple oscillator has to be made, then that oscillator has to be modified to make it voltage controlled. And at last, the signal out of the oscillator needs to be modulated before it can be outputted. Using operational amplifiers most of this can be done. With two opamps a sawtooth oscillator can be made; this oscillator can be made voltage controlled by using bipolar junction transistors as resistors. This has the extra benefit of creating a certain frequency at 1V / octave. To modulate the sawtooth, it can be inserted into an inverting amplifier with a gain equal to 1. This inverts the sawtooth. Then the inverted sawtooth can be inserted into a comparator that makes a square wave. Any of these three signals can be tapped and buffered to send to the output of the module. This is visualised in the diagram.

Design

On the left is the PCB design shown in altium, on the right is the board after all of its components have been soldered. The size of the board is chosen according to the eurorack standard, so it can be easily implanted into the casing of the synthesizer. The QR-code contains information about the electronical schematics.

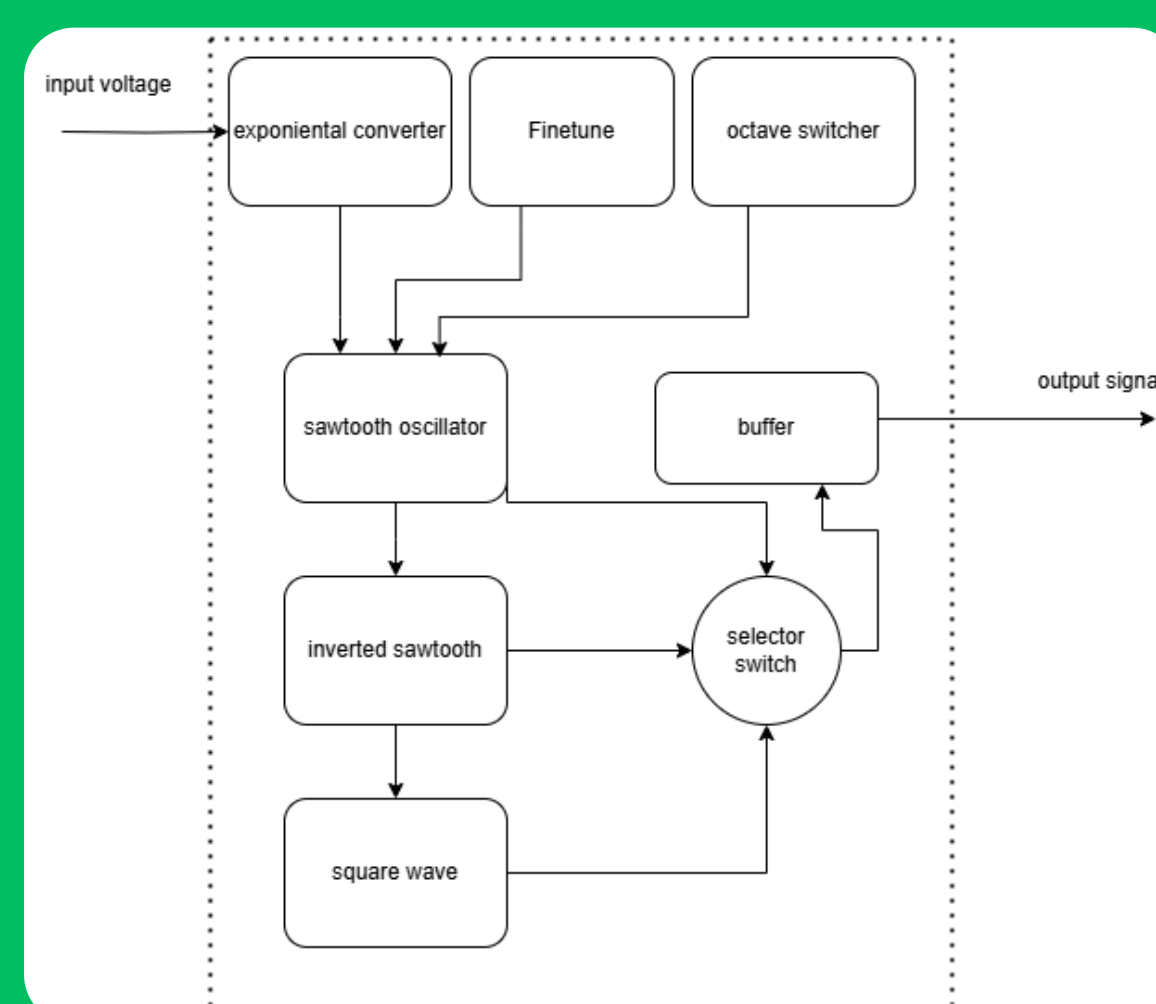
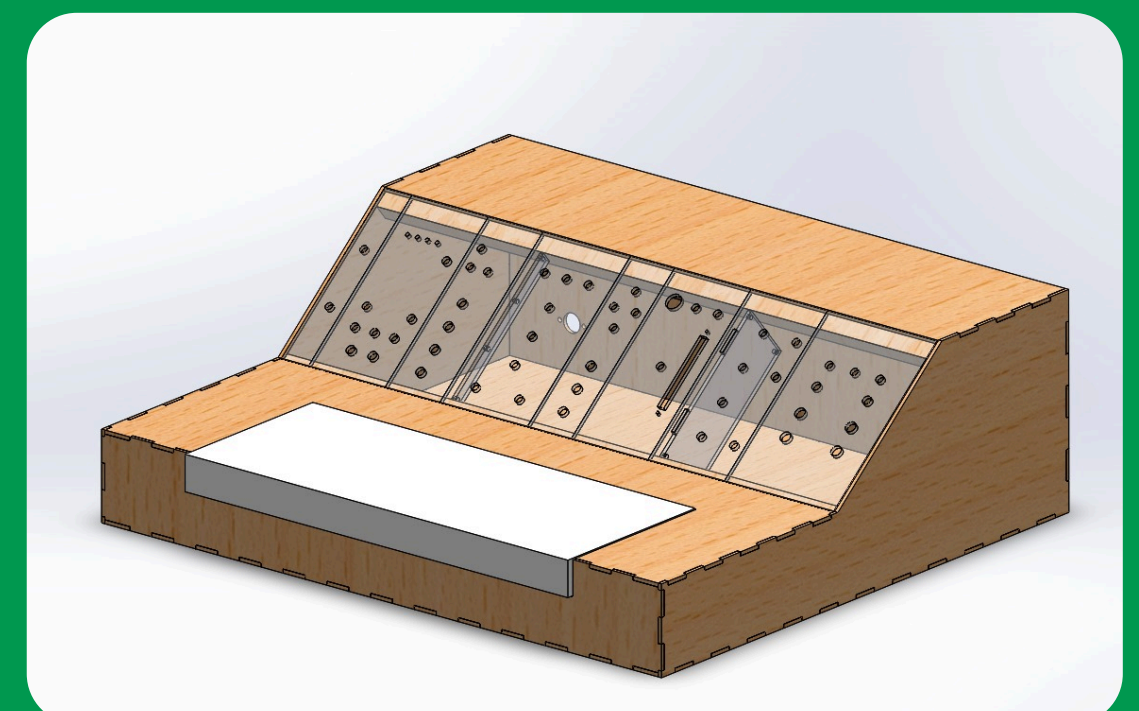


Results

The dual VCO module is operational. It's two independent channels can be switched from signal, octave layers and finetuned. Shown in the picture is a computer drawing of the synthesizer wich will house the module.

Conclusion

Although the end product has been made, the road to that point was not easy. The PCB design contained a couple of routing errors, resulting in some external routes that had to be added. Also, some resistors had to be changed to a higher value to limit current flow in the board. But all the (non) functional requirements have been met.



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